DEEP LEARNING FOR MALARIA DIAGNOSIS

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1. **ABSTRACT-** Malaria is a female Anopheles mosquito-borne disease that transmits a motile infective form to the host body such as humans, which reproduce asexually in the blood cells of the host. The typical symptoms of malaria are fever, headache, tiredness, and vomiting. In severe cases may lead to coma and death. In this research, we used deep neural networks to detect the malaria virus in human blood cells. Traditional malaria detection techniques require experts to test blood cells under a microscope. The shortage of skilled technicians and the unavailability of required equipment and infrastructure result in false diagnoses leading to an increase in mortality rate. Early researches have shown detection using Machine Learning techniques like Support Vector Machine (SVM), which is tedious and requires hand-engineered feature extraction to train data, and the results were not up to the mark. The proposed method in this re- search shows a system with end-to-end automated models using a deep neural network that performs both feature extraction and classification using blood smear cell images. The dataset used in this research was taken from the National Institute of Health (NIH) Malaria Dataset. Models are evaluated based on accuracy, precision, recall, and F1- score. Data Pre-processing techniques like data segmentation and normalization are applied to maximize the model performance. In this research, two deep Convolutional Neural Networks, VGG-19, and ResNet-50, are experimented on to analyze and compare the best performing model on the malaria dataset [25]. Our results showed that the ResNet-50 outperforms the VGG-19 by achieving an accuracy of 97 percent [15].

KEYWORDS: Deep Learning, Tensor flow, Convolution, Pooling, RelU, epoc

2. INTRODUCTION

Malaria, a mosquito-borne disease caused by the Plasmodium parasite, affects millions of people annually, resulting in significant mortality rates. Diagnosis traditionally relies on light microscopy, which faces challenges due to a shortage of skilled technicians in affected regions, leading to delays and inaccuracies in diagnosis. This inaccurate diagnosis contributes to higher mortality rates, drug resistance, and unnecessary economic burdens.

To address these challenges, the research focuses on designing an accurate malaria diagnosis model that eliminates the dependency on skilled technicians. The goal is to provide a reliable alternative for routine high-quality diagnosis, especially in areas lacking easy access to healthcare [8].

The proposed solution involves leveraging deep learning, specifically Convolutional Neural Networks (CNNs), for malaria diagnosis. Unlike traditional machine learning methods, deep learning models, such as RESNET-50 and VGG-19, are scalable, reliable, and exhibit higher accuracy rates. CNNs, in particular, have proven effective in various computer vision tasks.

The research aims to answer two key questions:

1. Model Comparison: Evaluate the performance of two CNNs, RESNET-50 and VGG-19, on the malaria dataset to determine which one performs better for diagnosis,
2. Effect of Preprocessing: Investigate the impact of preprocessing techniques, such as data augmentation, stain normalization, and standardization, on the performance of CNNs for malaria diagnosis.

Automated image analysis software is proposed to overcome the limitations of conventional microscopy, providing a faster and more efficient diagnosis process. The ultimate objective is to enhance early and accurate diagnosis, leading to prompt treatment, thereby saving lives and reducing the global burden of malaria.

3. LITERATURE SURVEY

3.1 Malaria and its Diagnosis

Malaria is a parasite-borne illness that is carried by mosquitoes and poses a severe danger to global health. An estimated 300–500 million cases of malaria occur annually,

with 1.5–2.7 million deaths, underscoring the importance of the disease [2]. Light microscopy-based conventional diagnostic techniques are acknowledged and entail the study of stained blood smears. However, there are problems with these methods, such as the dependence on skilled experts, which results in a delay in receiving an accurate diagnosis. This reliance on manual microscopy turns out to be inefficient and increases mortality rates in places where access to healthcare is scarce [5]. In this context, the significance of efficient and automated diagnostic tools is underscored, opening the door for further research into state-of-the-art techniques such as deep learning for malaria diagnosis.

3.2 Machine Learning Applications in Health Care

In the sphere of healthcare, the use of computer-aided diagnostic instruments powered by machine learning algorithms has largely supplanted conventional diagnostic techniques. This change was prompted by the prospect of increased accuracy and efficiency in medical diagnosis. Previous research endeavors have thoroughly studied a range of machine learning methods, including logistic regression, Bayesian learning, support vector machines, and k-nearest neighbor algorithms, to solve the issues associated with malaria detection. Despite striking results, challenges have been found with manually constructed feature extraction methods. Deep learning is being used more and more in the healthcare sector as a way to overcome these challenges because it is thought to be a more adaptable and efficient approach. With the promise of improved diagnostic efficiency and accuracy, machine learning (ML) applications have replaced traditional diagnostic approaches in the healthcare industry [7]. To overcome the difficulties in malaria diagnosis, a variety of machine learning techniques were investigated, including support vector machines, k-nearest neighbor algorithms, logistic regression, and Bayesian learning. On the other hand, the shortcomings of manually created feature extraction methods account for the failure in certain cases. The inability of these approaches to fully capture the intricacy of medical data resulted in less than ideal outcomes [16].

3.3 Deep Learning Applications in Health Care

Medical image analysis has greatly increased with the use of deep learning, specifically Convolutional Neural Networks (CNNs), which have improved diagnostic accuracy and automated difficult tasks [3]. This literature review acknowledges deep learning's broad use in the medical field and highlights how it may be used to assess pathology slides and X-rays, among other medical pictures. However, because deep learning models are black-box, making it difficult for physicians to understand that, questions have been raised about their interpretability and simplicity in medical settings [4]. Implementation must be done responsibly because of ethical issues, such as patient privacy and possible biases [6]. The goal of ongoing research is to overcome these issues and strike a compromise between upholding ethical norms, transparency, and trust while utilizing deep learning's efficiency in the healthcare industry [9].

4. MODEL SELECTION

4.1 VGG - 19

Oxford's Visual Geometry Group produced a 19-layer deep Convolutional Neural Network (CNN) called VGG-19, which is a variation of the VGG model [23].

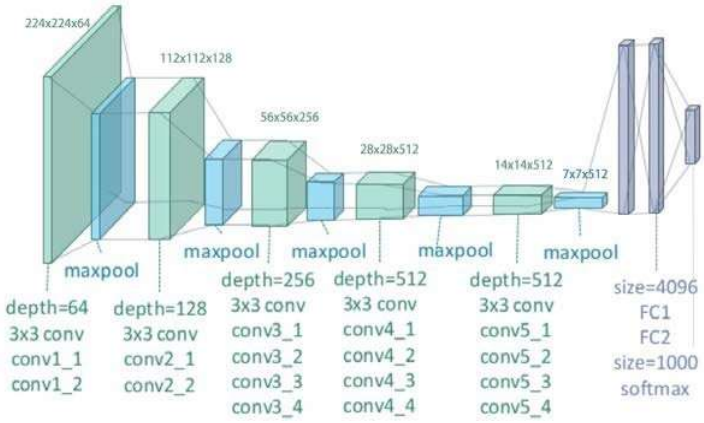

Fig1: VGG-19

4.2 Transfer Learning

The transfer learning, which enables to apply the knowledge of pre-trained models to their own problem sets [42]. Rather than beginning from zero, we leveraged models built on large-scale datasets such as Image-Net (100,000 data-points). This study highlights the effectiveness of transfer learning, which has been shown to be beneficial in a number of picture categorization studies [10]. In particular, we used CNNs as feature extractors that had been pre-trained on extensive datasets like VGG-19 and ResNet-50. When compared to conventional techniques, this approach performs better since it recognizes the models' capacity to generalize information from a variety of datasets [11].

4.3 ResNet-50

Deep neural network training was revolutionized when VGG-19, a Convolutional Neural Residual Network (ResNet), appeared as a key answer to the vanishing gradient problem. ResNet, the winner of the 2015 ImageNet competition, improved results and lessened the gradient

problem with its skip connection strategy, paving the way for models such as ResNet50. Using ResNet50 and transfer learning, this study freezes the first blocks and uses fine-tuning in addition to preprocessing methods. The five stages of the ResNet-50 model, which has 25 million trainable parameters, each include identity and convolution blocks.

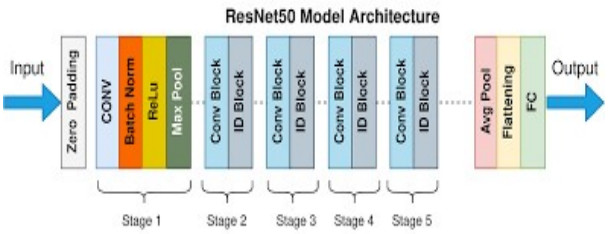

Fig 2: ResNet-50

For binary image classification, sigmoid activation is preferred over soft-max. Accuracy and Loss are two evaluation metrics that are assessed in 64-sized batches utilizing the RMS Prop-optimizer and binary cross-entropy during training and validation.

4.4 Keras

It is a Python module for deep learning that operates on top of the Tensor Flow library

5. PROPOSED MODEL

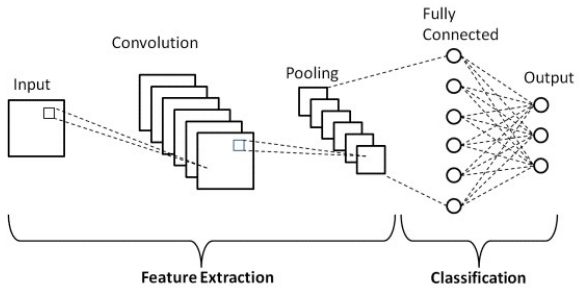

Fig 5.1: Model

5.1 Objective

This proposed Convolution Neural Network (CNN) model aims to address the problem of image classification, where the goal is to classify image into predefined categories [1]. The primary objectives are to improve classification accuracy, reduce over-fitting, and enhance the model's ability to capture intricate patterns in images [18].

5.2 Architecture Overview

The proposed CNN consists of three convolutional layers, each followed by a Rectified Linear Unit (ReLU) activation function. Max pooling is applied after each convolutional layer to downsample the feature maps. The final layer is a fully connected layer with softmax activation for classification [12], [13].

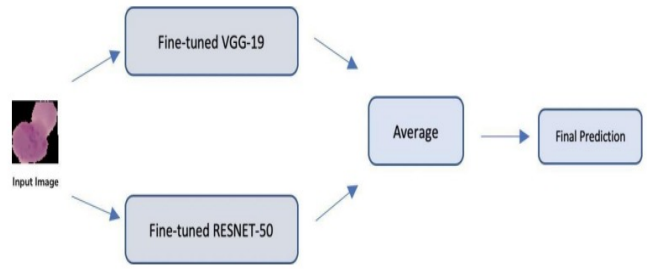

Fig 5.2: Model approach

5.3 Key Components

An attention mechanism is integrated after the second convolutional layer to enhance the model's focus on important regions. This attention mechanism allows the model to prioritize relevant features during the learning process [14].

5.4 Hyper-parameters

Key hyper-parameters in ResNet-50 include a learning rate of 0.001 and set $\mathbf{lr = 1e - 4}$, a batch size of 64, and dropout with a rate of 0.3 to prevent over-fitting [20]. These values were selected through experimentation and grid search [26].

5.5 Training Procedure

The model is trained on a diverse dataset of **27,557** images, of which 13,778 are parasitized images and 13,779 are non-infected images. We merged files of both non-infected images and parasitized images and labeled them as healthy and malaria for the respective files [11]. The dataset is divided into three sets for training, validation, and testing [17].

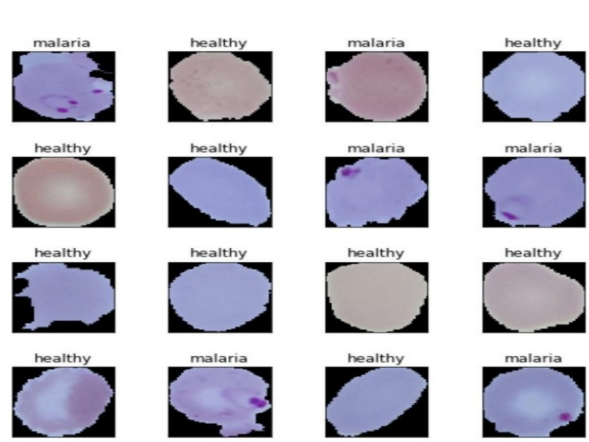

Fig 5.3: Malaria Dataset

The ratio of train to validation to test is 63:7:30. We choose to take only 63% of dataset as training to avoid high computational intensity due to large dataset [24], [22].

5.6 Validation and Testing

The model's performance is evaluated on a separate validation set, and the metrics include accuracy, precision, recall, and F1 score [21]. A test set, not seen during training,

is used for final evaluation[14].

6. RESULT ANALYSIS

Using transfer learning and fine-tuning on a malaria dataset, pre-trained Convolutional Neural Networks (CNNs) are employed in the study of human red blood cell smear pictures to identify them as healthy or infected [19]. Results show that preprocessing methods such as scaling, stain standardization, and normalization have little effect on the performance of VGG-19 and ResNet-50. On the other hand, data augmentation shows encouraging outcomes. ResNet-50, on the other hand, outperforms VGG-19, which achieves 96.24% accuracy after fine-tuning, with a noteworthy 97.09% accuracy with data augmentation. Transfer learning works well, demonstrating the superiority of ResNet-50 with an accuracy of 97.09%. An ensemble model obtained 96.76% accuracy, compared to the previous state-of-the-art ResNet accuracy of 95.7%.

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Model Performance metrics:
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Accuracy: 0.9624
Precision: 0.9629
Recall: 0.9624
F1 Score: 0.9624

Model Classification report:
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              precision    recall  f1-score   support

     healthy       0.95      0.98      0.96      4144
     malaria       0.98      0.95      0.96      4124

    accuracy                           0.96      8268
   macro avg       0.96      0.96      0.96      8268
weighted avg       0.96      0.96      0.96      8268


Prediction Confusion Matrix:
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                 Predicted:
                    healthy malaria
Actual: healthy        4059      85
        malaria         226    3898
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Fig 6.1: Fine-tuned model for both VGG-19 and ResNet-50

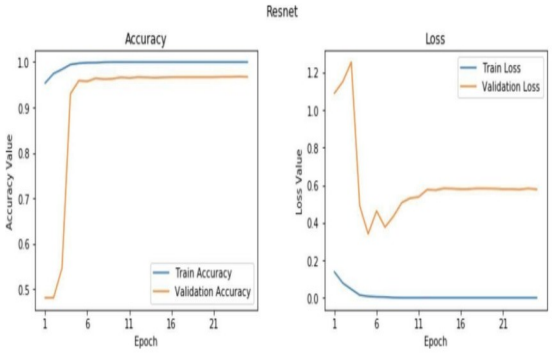

Fig 6.2: Accuracy plot ResNet-50

Model	Accuracy (Test)	Precision	Recall	F1 score
VGG-19 basic	93.21	.9322	.9321	.9321
VGG-19 fine tuned	96.24	.9629	.9624	.9624
ResNet-50	97.09	.9709	.9709	.9709

Fig 6.3: Model Performance

7. CONCLUSION

In this paper, we investigated how well end-to-end deep learning neural networks can improve the classification of malaria diagnoses. Preprocessing methods such as staining, normalization, and standardization did not significantly affect performance, according to our research. On the other hand, data augmentation produced favorable results that enhanced the performance of the model. VGG-19 and ResNet-50 were compared, both built from scratch and via transfer learning with hyper-parameter adjustment. The results showed that ResNet-50 performed better, with an accuracy of 97.09%. We will now turn our attention to network architecture optimization in order to pinpoint important performance aspects and improve model performance via more testing and hyper-parameter tuning.

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